Handout Title

Spring 2026 · Course Name

1 Energy and Conservation

A particle of mass m moves in a one-dimensional potential $V(x)$. The total mechanical energy is

$$E = \frac{1}{2}mv^2 + V(x), \tag{1}$$

which is conserved when no external forces act.

Definition 1.1. A force is *conservative* if it can be written as the negative gradient of a potential, $F(x) = -\,\mathrm{d}V/\mathrm{d}x$.

Theorem 1.2. *For a conservative force, the total mechanical energy E is constant along any solution of the equation of motion.*

Remark 1.3. The converse fails in general: dissipative systems admit no such potential, and E is not conserved.

Exercise 1.1

Consider a particle of mass m in the potential $V(x) = \frac{1}{2}kx^2$.

a) Write down the equation of motion using Newton's second law.

b) Show that $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$ is conserved along solutions.

The result can be verified by differentiating E with respect to time and using the equation of motion from part (a).

c) Find the angular frequency of small oscillations.

Exercise 1.2

A block of mass m slides down a frictionless inclined plane that makes an angle θ with the horizontal.

a) Find the acceleration along the plane.

b) How long does it take to slide a distance L from rest?

c) What is the speed at the bottom?

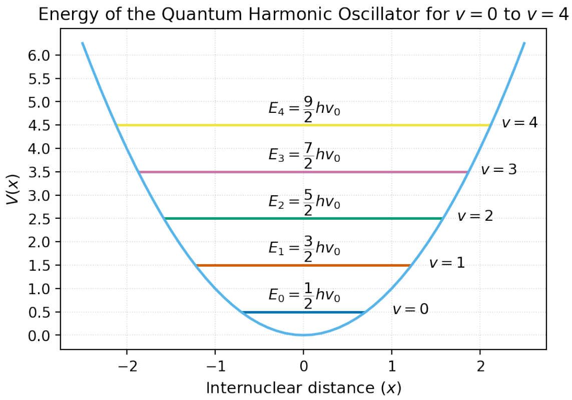

Figure 1: Schematic figure for the topic.

2 Circular Motion

In uniform circular motion, the centripetal acceleration has magnitude $a_c = v^2/r$, directed toward the center of the circle.

Exercise 2.1

A satellite moves in a circular orbit at radius r around a planet of mass M.

a) Show that the orbital speed is $v = \sqrt{GM/r}$, where G is Newton's gravitational constant.

b) Express the orbital period T in terms of r, M, and G.